Deniable Secret Sharing: Towards Practical Faking

Ignacio Amores
Aarhus University

Tamar Ben David
Ariel University

Matteo Campanelli
Offchain Labs and University of Tartu

Sebastian Kolby
Aarhus University

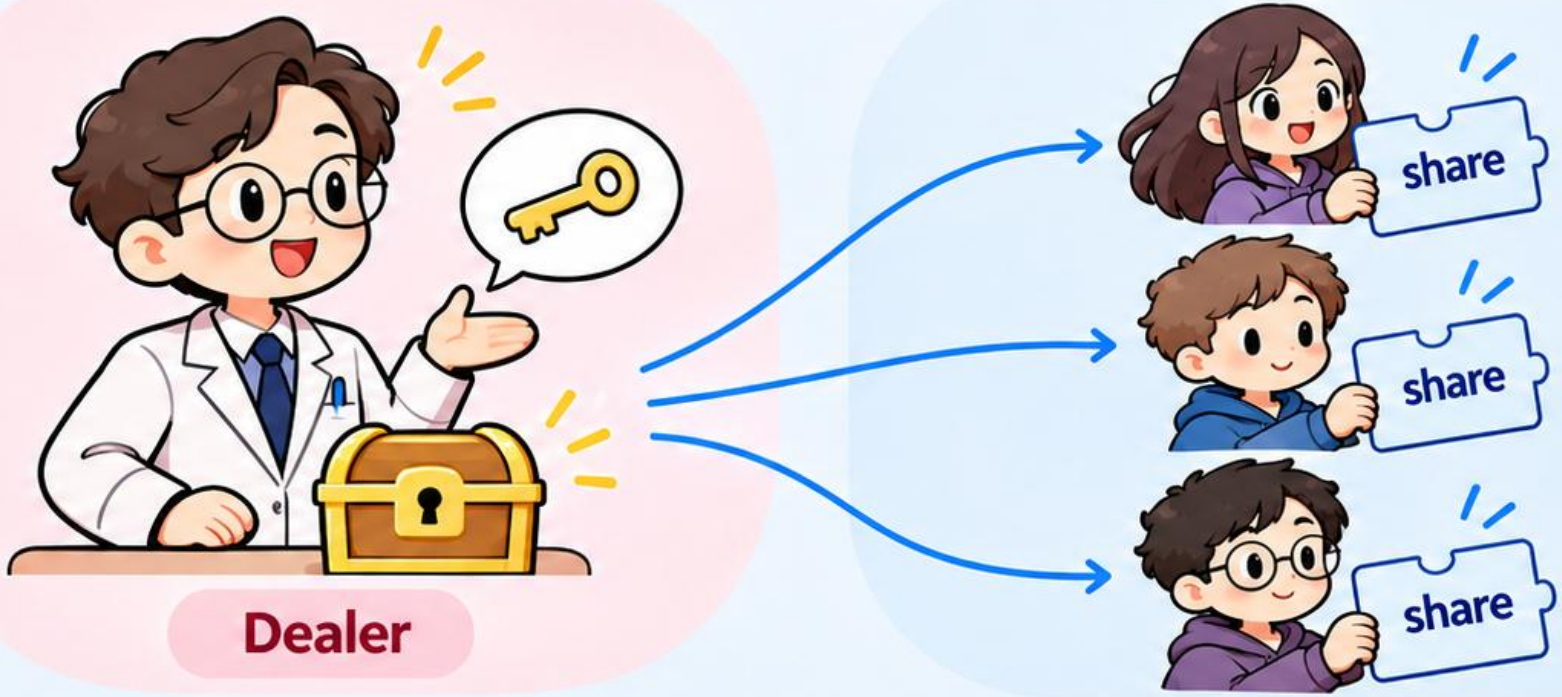
Eran Omri
Ariel University

Anat Paskin-Cherniavsky
Ariel University

Sophia Yakoubov
Aarhus University



Standard Secret Sharing



Shareholders / Players

Share

Rec

2-out-of-3 Shamir example



can reconstruct

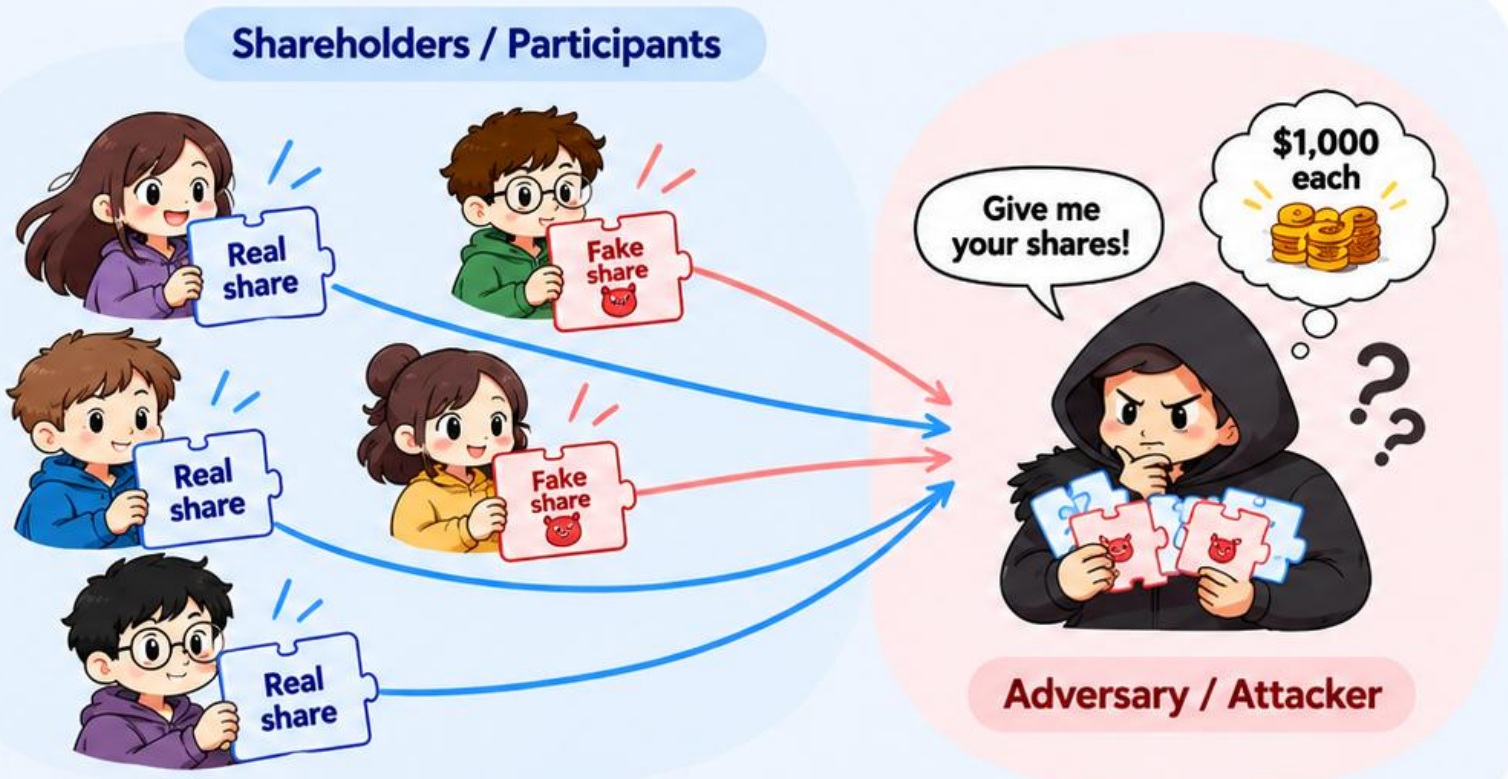


cannot reconstruct

Ran Canetti • Ivan Damgård • Sebastian Kolby • Divya Ravi • Sophia Yakoubov

What is deniable secret sharing?

DSS protects secret sharing against bribery and coercion.



SHARE

REC

FAKE

DSS: Fakers vs. Snitches

Shareholders may respond differently to the adversary's coercion.



Snitches

Hand over their true shares.



Fakers

Follow the faking mechanism and produce fake shares.

Privacy should hold as long as the snitches are unqualified.

Stronger Security Properties of DSS

Variant Privacy

Fake-Real Indistinguishability (FRI)

Original shares

Sent to adversary

$$\begin{bmatrix} m_0 & m_0 \\ m_1 & m_0 \end{bmatrix} \overset{\text{FRI}}{\approx} \begin{bmatrix} m_1 & m_1 \\ m_1 & m_0 \end{bmatrix}$$

Faker

Snitch

Fake Hiding (FH)

Original shares

Sent to adversary

$$\begin{bmatrix} m_0 & m_0 \\ m_1 & m_0 \end{bmatrix} \overset{\text{FH}}{\approx} \begin{bmatrix} m_1 & m_1 \\ m_1 & m_1 \end{bmatrix}$$

The Challenge: Practical Faking

Design a faking mechanism that remains secure even when some shareholders snitch.



Message Complexity

Visible communication may raise suspicion.

Minimize messages



Privacy Against Fakers

Unqualified sets may learn the secret.

Preserve privacy

Non-Interactive Faking

No online communication.

Information-Theoretic Limitations

Perfect Fake Hiding (FH)

1 **Non-zero communication is necessary**



2 **For linear schemes, privacy against fakers is lost**



Why Is Faking Difficult? — Shamir Example

Perfect Fake Hiding

2-out-of-3 threshold

Faker A

$p(1), p'(1)$

Faker B

$p(2), p'(2)$

Snitch C

$p(3) = p'(3)$

Difference polynomial

$$q(x) = p'(x) - p(x)$$

Faker A knows

$q(1)$

At the snitch's point

$q(3) = 0$

↓

Two points determine a line

$$m = m_{\text{fake}} - q(0)$$

Theorem 2: General Lower Bound (Information-Theoretic)

$$MC_{FH} \geq \tau(G)$$

MC_{FH}: minimum number of messages exchanged in a faking protocol that achieves Fake Hiding.

$\tau(G)$: size of a minimum vertex cover of the hypergraph G.

2-out-of-3 Shamir (one snitch)

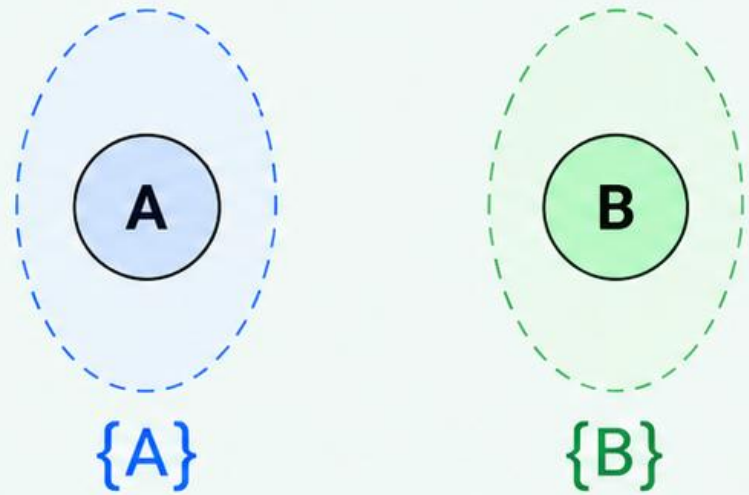


Minimally qualified sets (size 2):

{A, B} {A, C} {B, C}

Keep sets with the snitch and remove the snitch

Hypergraph G (vertices are the fakers)



Theorem 7: Achieving FRI with Exponential Share Size – PRG

$$\mathrm{PRG} : \{0,1\}^{\lambda} \to \{0,1\}^{\lambda} \times \{0,1\}^{\ell_{share}}.$$

Share(m)

$(s_i)_{i\in[n]} \leftarrow \mathrm{S.Share}(m)$
for $F \in 2^{[n]}$:
    $r_F \leftarrow \{0,1\}^{\lambda}$
    for $i \in F$:
        $R_i[F] \leftarrow r_F$
return $(s_i, R_i)_{i\in[n]}$

Fake$(F, m^{\mathrm{fake}}, R_i)$

$R'_i \leftarrow R_i$
$r_1 || r_2 \leftarrow \mathrm{PRG}(R_i[F])$
$R'_i[F] \leftarrow r_1$
$(s'_i)_{i\in[n]} \leftarrow \mathrm{S.Share}(m^{\mathrm{fake}}; r_2)$
return $(s'_i, R'_i)_{i\in[n]}$

Theorem 8:
Achieving FRI with Polynomial Share Size — Ideal VBB Obfuscation

New Construction

$$\widetilde{\mathrm{Rand}}(c_i, F)$$

Generate shared randommess on demand

Polynomial Share Size

Previous Construction

$$\{r_F\}_{F \subseteq [n]}$$

Store randommess for every possible faker set

Exponential Share Size

FRI + Non-Interactive Faking

Assumption: Ideal VBB (Oracle-Access Model)

Open Direction: Replace Ideal VBB with Weaker Assumptions (e.g., IO)

Open Questions



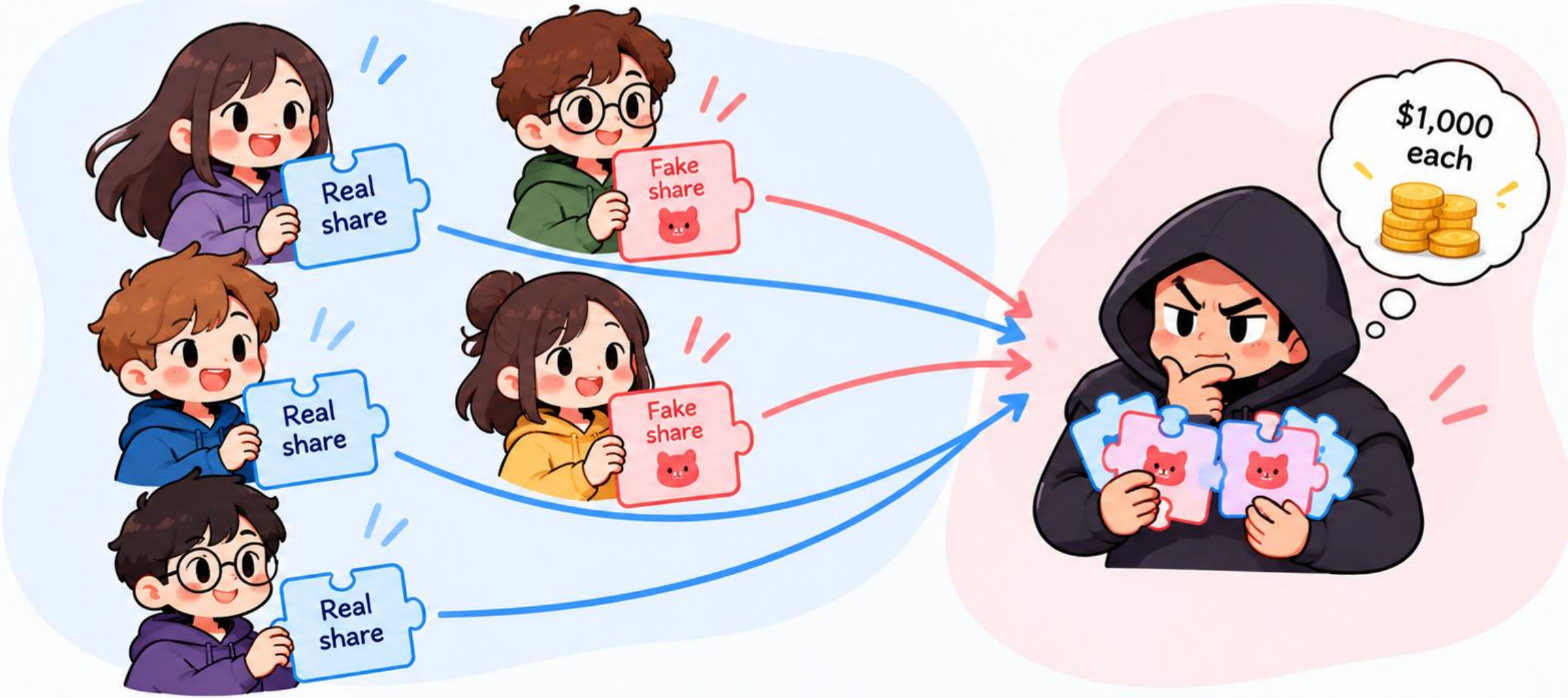


Computational Setting



Information-Theoretic Setting

Thank you!


Real share
Fake share
Real share
Fake share
Real share
$1,000 each